higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

COMPUTER PROGRAMMING

(Second Paper)

NQF LEVEL 4

(10041024)

14 March 2019 (Y-Paper)
13:00–17:00

This question paper consists of 13 pages.

NCV

TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.

INSTRUCTIONS TO STUDENTS AND LECTURERS

1. Students MUST be seated in the computer laboratory 30 minutes before the start of the examination to ensure that ALL computers have the appropriate programs.
2. These questions must be completed on a computer by making use of MS Office 2010 or a later version and VB.NET 2005 or VB.NET Express or VB.NET 2008 or a later version.
3. Students must make sure that they print their work immediately after completing a question. Ensure that your EXAMINATION NUMBER appears on EACH printout. NO questions without a printed EXAMINATION NUMBER will be marked.
4. In the event of problems such as a power failure, computer breakdown or a printer breakdown, the invigilator will make the necessary arrangements to continue with the examination process once the problem has been resolved. Students will not be penalised with time lost under these circumstances.
5. All work on the computer should be saved at regular intervals to prevent loss of work. NO additional time may be allowed for work lost due to lack of saving.
6. NO student may print his/her work for another student, or make a memory stick available to another student or access another student's work on the network. Any attempt to access any information from or transfer information to another student in whatever manner is a contravention of the examination rules and regulations and will be viewed in an extremely serious light.

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7. ALL PRINTOUTS or screen prints done during the session MUST be handed in.

8. Make sure that your examination number appears on all printouts.

9. Students may NOT copy the source code from VB.NET to MS Word and then print it from the word processor.

10. Steps to print a VB.NET form or any other screen:

    10.1 Run the solution.

    10.2 Hold the ALT key down and press the print screen button (This will copy the active screen).

    10.3 Open MS Word and paste the form/screen in a new document.

    10.4 Add your examination number to the top right-hand side of your page.

    10.5 Print the page.

11. You are allowed to make use of the help files in VB.NET.

12. The following files will appear on your hard drive in the folders:

    12.1 QUESTION 1
    C:\CP L4 P2\CarEvaluator
    CarEvaluator (Microsoft Visual Studio Solution)

    12.2 QUESTION 2
    C:\CP L4 P2\Databases\ CBT_Systems.accdb
    C:\ CP L4\ CBT_SYSTEMS
    CBT_SYSTEMS (Microsoft Visual Studio Solution)

    12.3 QUESTION 3
    C:\CP L4 P2\QUESTION 3
    Images\survey.jpg
    Style1.css

    12.4 QUESTION 4
    C:\CP L4 P2\QUESTION 4
    Index.html

13. Students may use the following or any other application of their choice to create the specified webpages:

    Notepad ++, NetBeans, BlueVoda, Dreamweaver

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QUESTION 1

The following questions refer to the VB.NET solution CarEvaluator, which can be found in C:\CP L4 P2\CarEvaluator on your hard drive.

You are requested to create an application which will help Supercar to evaluate customer cars and calculate their resale price. The program captures the make of the car, purchase price and mileage (i.e. the total number of kilometers on the car's odometer).

Below is the IPO chart for the problem.

INPUT	PROCESSING	OUTPUT
Make purchaseprice mileage	Determine the resaleprice: resaleprice = purchaseprice - (purchaseprice x percentage)	Make purchaseprice mileage resaleprice

FIGURE 1 below is a print screen of what the user interface looks like.

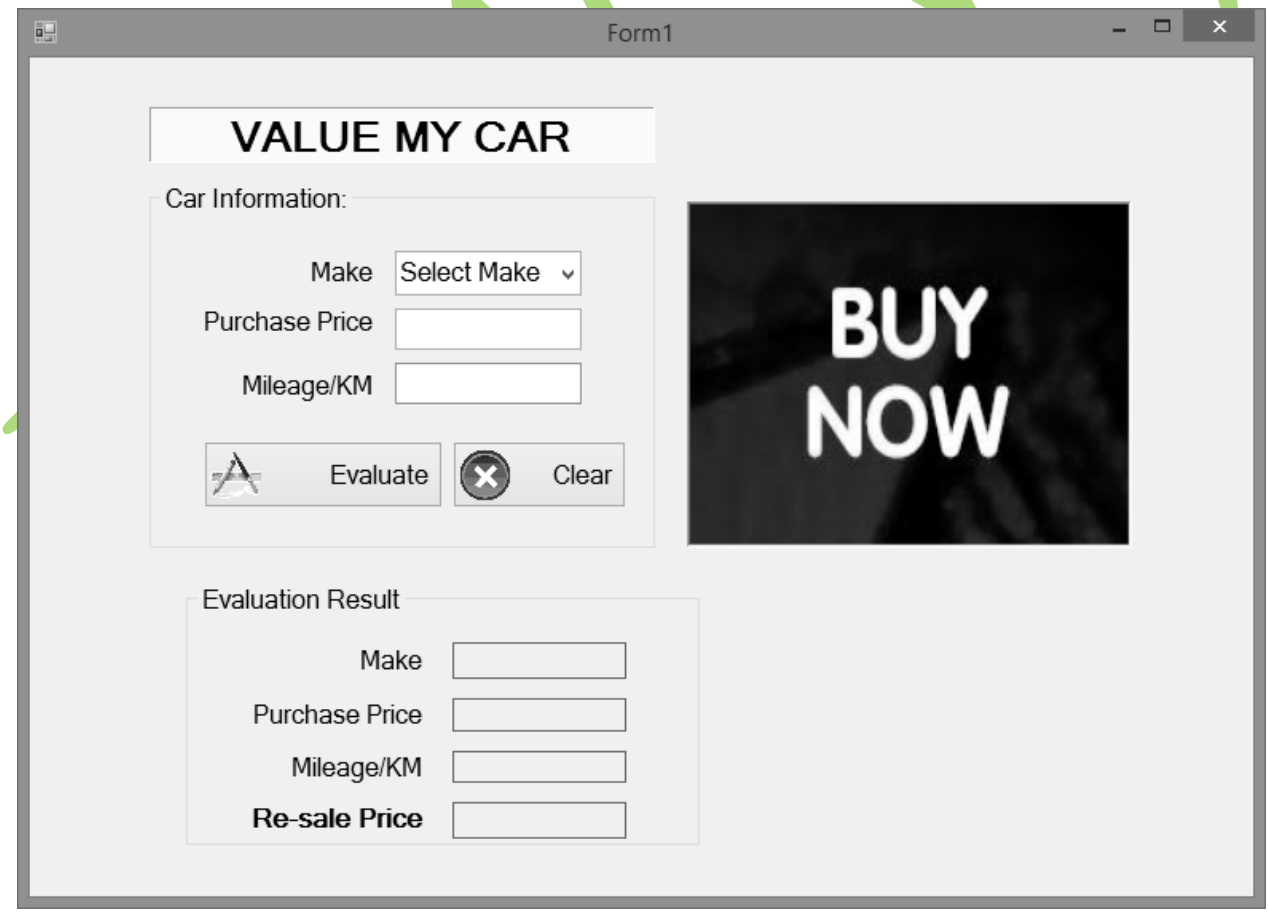


FIGURE 1

The following STEPS explain how to use the program to evaluate a car:

Step 1: Run the program, select the car make and enter the purchase price and mileage:

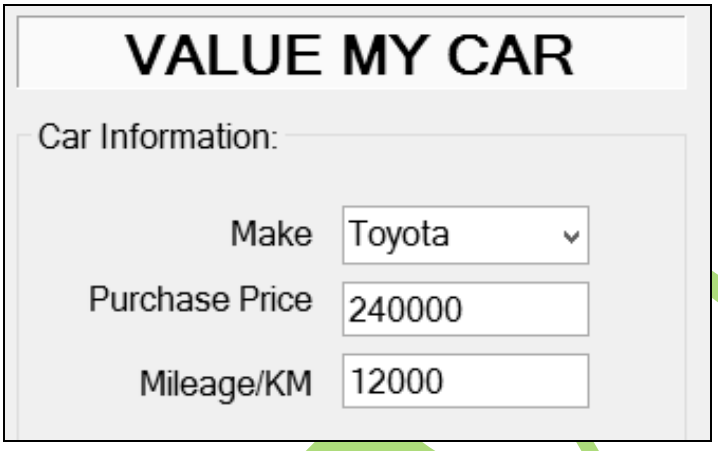


FIGURE 2

Step 2: Click on the process evaluate button to calculate the resale price.

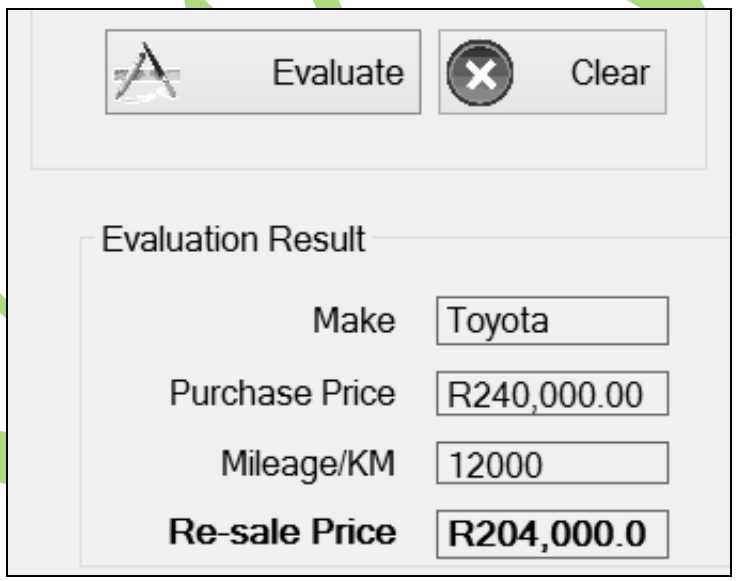


FIGURE 3

Study the following class diagram to help you complete the specified tasks.

EVALUATE
-strMake -dblPurchaseprice -intMileage -decPercent
+calResale ()

The completed interface with partially completed program code has been saved on the hard drive. Open the solution and follow the instructions to complete the program.

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1.1 Add your EXAMINATION NUMBER to the program code of the clsCarEvaluator as a comment and complete the following.

1.1.1 Declare ALL the variables of the *Evaluate* class. (2)

1.1.2 Add a public read-write property called *mile* to the *Evaluate* class that will calculate and return the car's de-valuation percentage based on the kilometers travelled. Use a *case select* decision structure to determine the devalue percentage of the car as per the following data range: (7)

KILOMETERS	**PERCENTAGE**
10000 to 30000	15%
40000 to 99999	35%
100000 to 199999	60%
220000 to 400000	75%

1.1.3 Add a read-only public property called *percentage* to the *Evaluate* class that will return the calculated *percent* value. (3)

1.1.4 Add a public method called *calResale*() to the *Evaluate* class to calculate the resale price of the car. (4)

Make a PRINTOUT of the source code of the class.

1.2 Add your EXAMINATION NUMBER to the main form of the program. Add the following program code to this form:

1.2.1 Instantiate a new object called *objEvaluate* of the class *clsCarEvaluator* (2)

1.2.2 Assign the values from the *cmbmake* and *txtpurchase* variables to the appropriate properties of the *objEvaluate* object. (3)

1.2.3 Invoke the *calResale()* method of the class to calculate and display the resale price of a car. (2)

Display the output neatly in the appropriate controls. (See STEP 2 above.)

Add your EXAMINATION NUMBER to the program code of the main form, *frmCarEvaluator*.

Make a PRINTOUT of the program code of the form.

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1.3 Change the text in the title bar of the *frmCarEvaluator* form to your EXAMINATION NUMBER.

Run the program and enter the following test data. Click on the "Evaluate" button to display the results.

Make a PRINTOUT of each of the displayed outputs.

MAKE	PURCHASE PRICE	MILEAGE
BMW	R298 060,00	275 680
Toyota	R124 058,00	175 090

(2)
[25]

QUESTION 2

The following questions refer to the VB.Net solution called CBT_SYSTEMS (C::\CP L4 P2\ CBT_SYSTEMS that has been saved on the hard drive. A copy of the required database file has been saved in the following folder:
….\CP L4 P2\Databases\ CBT_Systems.accdb

In essence, the application represents a database driven computer based training system (CBT). Students are able to register new accounts, book assessments as well as view a list of scheduled assessments.

Below (FIGURE 4) is a graphical representation of the three data tables used by the application - *Assessment*, *BookAssessment* and *Register*. The relationship between the tables is also indicated on the diagram.

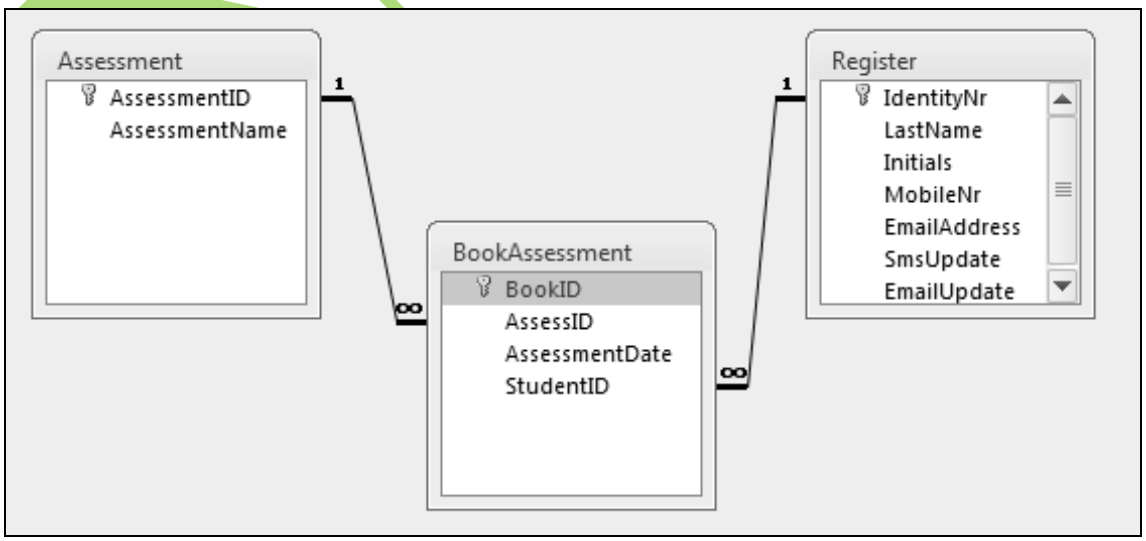


FIGURE 4

The following STEPS explain how to use the functionality of program to for example register a new account and book an assessment.

STEP 1 Register a new Account (*Register* table): Run the program. Click the “Create Account” tab page and complete the details as indicated in FIGURE 5 below. To save the data click the “Create Account” button.

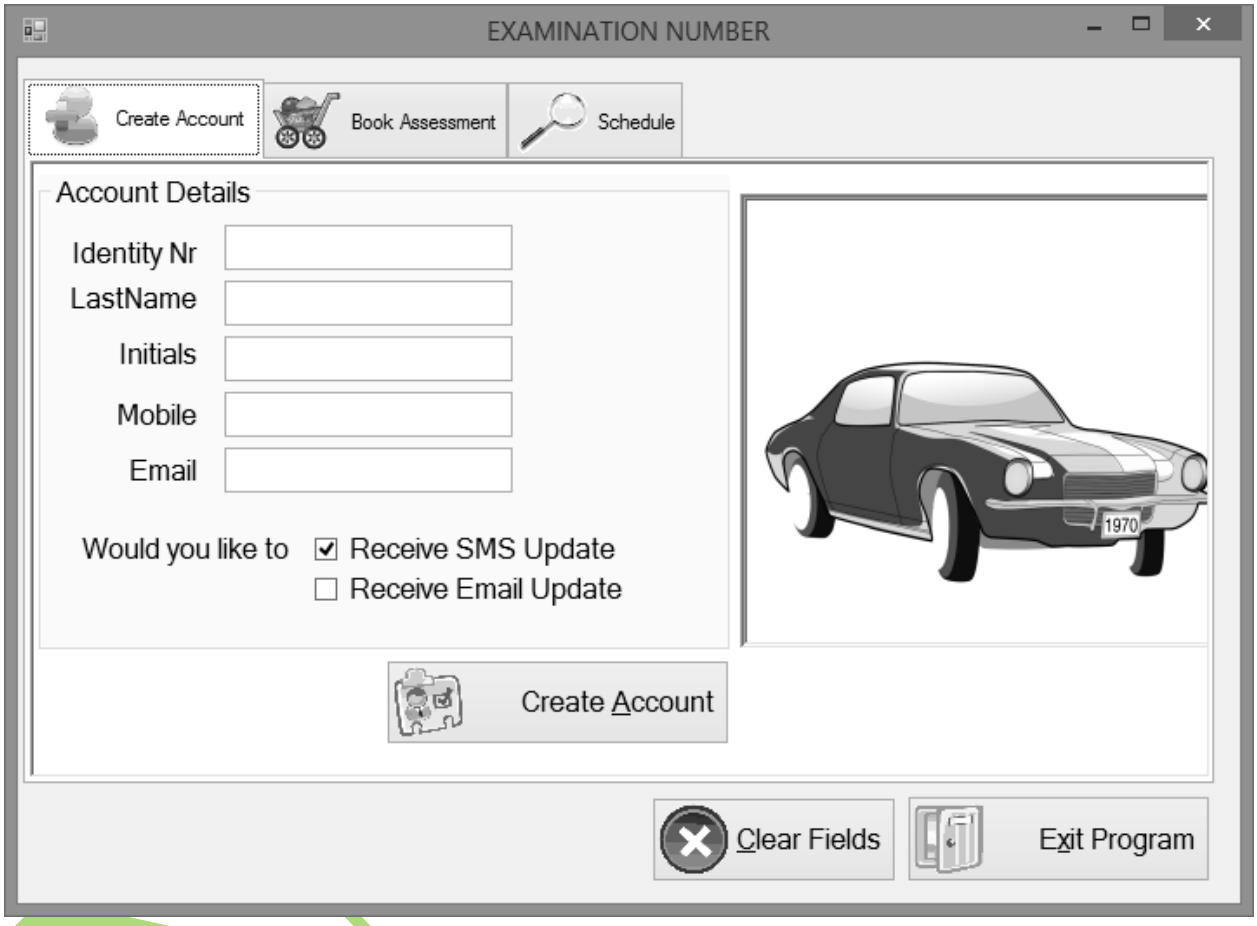


FIGURE 5

STEP 2 To book an assessment (*BookAssessment* table): Click the "Book Assessment" tab page. Enter the student identity number and complete the rest of the details as indicate in FIGURE 6 below. To confirm the booking click on the "Booking" button.

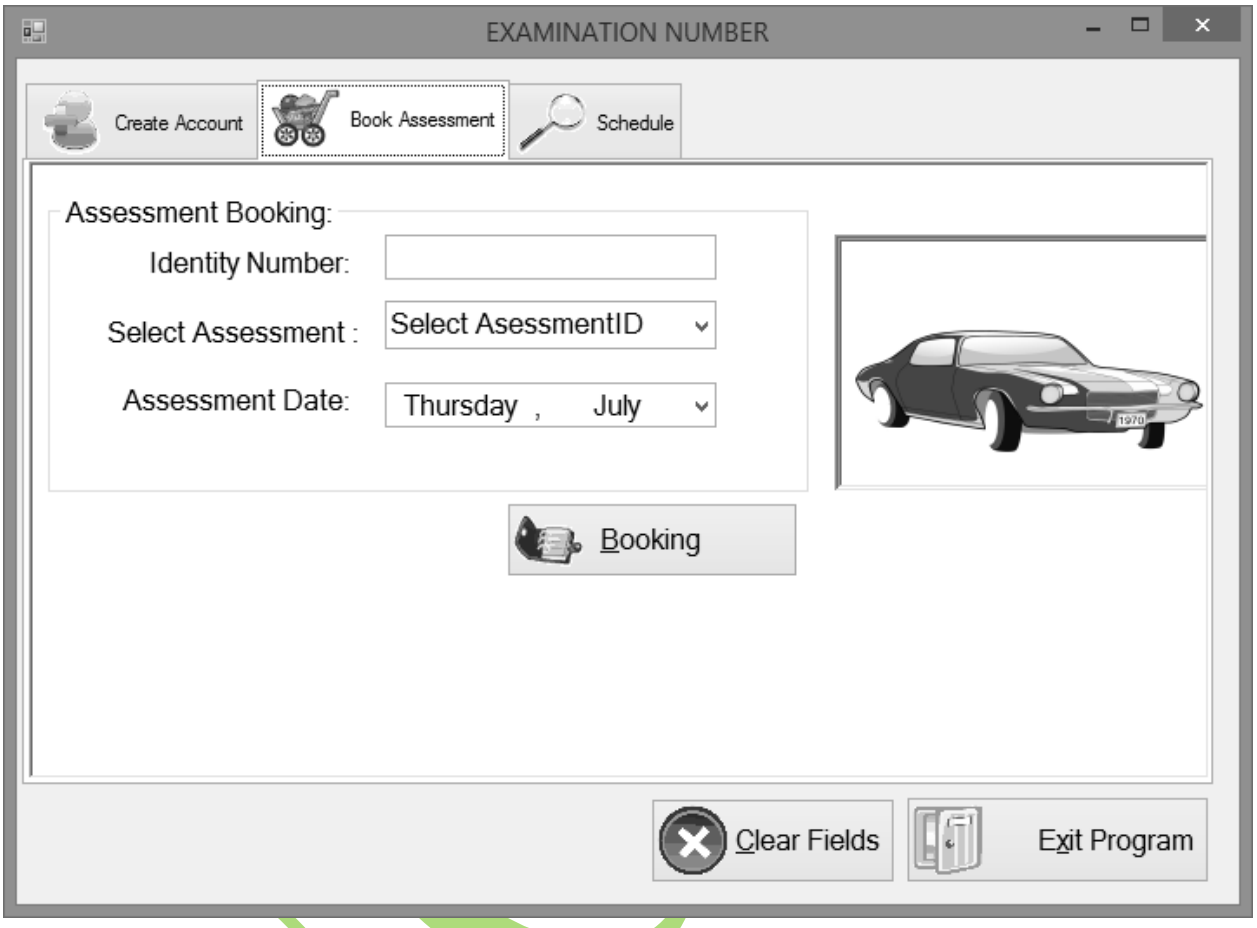


FIGURE 6

The completed interface with the partially completed program code has been saved on the drive disk. Open the solution *CBT_Systems* and follow the instructions below to complete the program:

NOTE: You are required to write the program code which contains SQL statements to complete the question.

2.1 Add the following program code to the getBooking() Sub Procedure. Add a *try … catch t*o the program code to ensure that any exceptions that might occur will be trapped and an appropriate error message is displayed to inform the user of the error.

2.1.1 Write the program code to establish a connection to the database. (3)

2.1.2 Write the SQL statement that will insert the data into the *BookAssessment* table when the student books an assessment. (10)

2.2 Add the code to the click event of *btnbooking* that will execute the *getBooking()* Sub procedure. (1)

2.3 Add the following program code to the *getSchedule()* Sub procedure:

Write the SQL statement that will retrieve the records from the *Register* and BookAssessment tables to display the student's schedule using the *studentID* as the filter parameter. (7)

2.4 Populate the *DataGridView* with the resulting records. (1)

2.5 Add the code to the click event of *btnSearch* that will execute the *getSchedule()* Sub procedure. (1)

2.6 Change the text in the title bar of the main CBT_Systems to your EXAMINATION NUMBER.

Run the program and enter the following test data. Make a PRINTOUT of each output result.

Student ID
0012250700082
9602147850820

(2)
[25]

QUESTION 3

You have been tasked to create a webpage for the WBE survey. The purpose of the webpage is to collect information from students about their WBE exposure and experience they have acquired.

NOTE: Create the home page (index.html) in QUESTION 3 and continue to QUESTION 4.

Below (FIGURE 7) is a screen print of the home page (index.html):

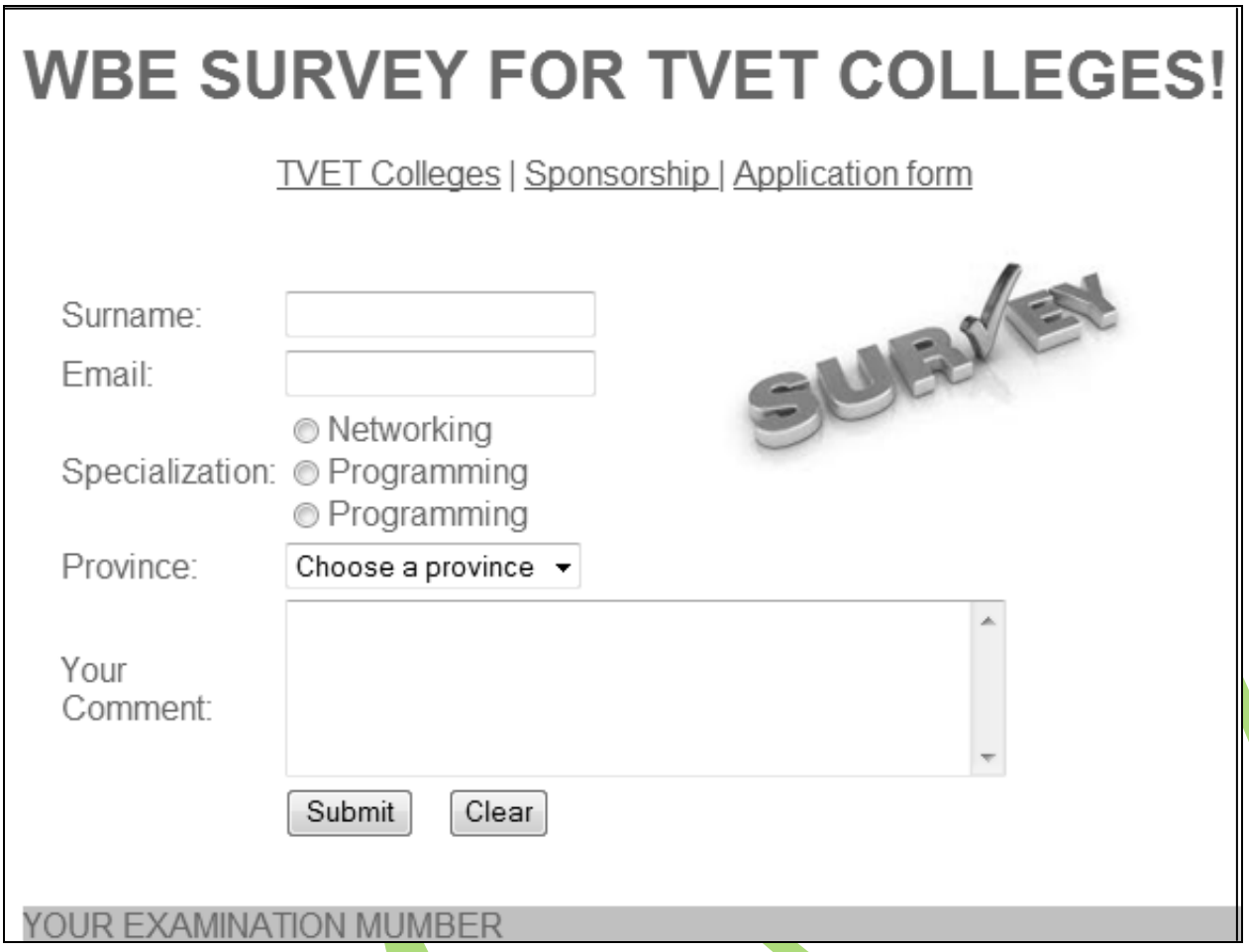


FIGURE 7

3.1 Open the external cascading style sheet called style1.css located in C:\CP L4 P2\QUESTION 3. Add your EXAMINATION NUMBER as a comment in the style sheet.

3.1.1 Create a rule set for the navigation division. Set the height to 20px, width to 570 pixels, text to align centre and a padding of 0px 15 px. (2)

3.1.2 Create a rule set for the right-side division. Set the position to absolute, 'left' to 320px and 'right' to 20px; (2)

3.1.3 Add a rule set which will remove the underlining from all the links. (1)

3.1.4 Add a rule set for a hover with a background colour of gray. (2)

3.1.5 Add a rule set for a footer in which the text is centered. (1)

Make a PRINTOUT of the CSS code.

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3.2 Create an HTML file and save it as Index.html. Insert your EXAMINATION NUMBER as the title of the page.

Add a link to the external style sheet file called Style1.css.

3.2.1 Define the container division with five div elements. The first element is the header and it will contain the heading 'TVET COLLEGE WBE' as in FIGURE 8. (2)

3.2.2 Make use of heading level 1 for the webpage heading 'TVET COLLEGE WBE'. (1)

3.2.3 Create the navigation division. Insert the hyperlinks TVET Colleges.html, Sponsorship.html and Application_Form.html. (3)

3.2.4 Create the right-side division. Insert the image saved in the folder Question 3\images\survey.jpg with the width of 230px and height of 150px. (3)

3.2.5 Create the content division. Insert a form element and a borderless table. (2)

3.2.6 Use the image in Figure 7 to help you add the following controls on the form element:

(a) TWO textboxes for user input
(b) TWO radio buttons for optional selection
(c) TWO buttons for *calculate* and *clear*

(3 × 2) (6)

3.2.7 Add a drop down list that contains the 9 provinces of South Africa. (5)

3.2.8 Add the 2 buttons to the form for *Submit* and *Clear.* (2)

3.2.9 Create the footer division. Insert your EXAMINATION NUMBER as the content footer of the division. (1)

Make a printout of the source code.

Make a PRINTOUT of the webpage as it appears in your web browser.

[33]

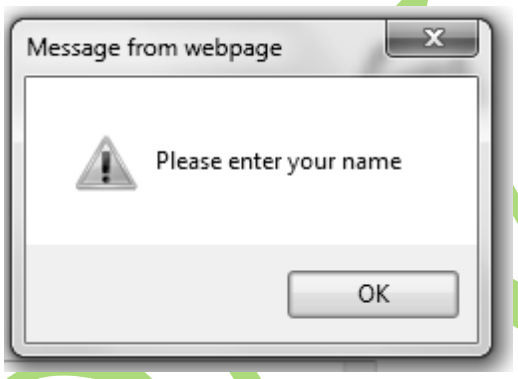
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QUESTION 4

Below is a screen print of the browsed webpage (*index.html*) which contains JavaScript.

4.1 Add a script element to the webpage and indicate that JavaScript will be used. (1)

4.2 Create a *JavaScript* function *called survey* to validate the surname and e-mail text boxes for content. if they are empty then the user must be prompted to enter the required information.



(8)

4.3 Provide the code that will determine if a province has been selected from the drop down list of provinces.

(4)

4.4 Display an appropriate message if no province has been selected and apply a border style of 2 pixels in solid red.

Make a PRINTOUT of the source code and print the webpage as it appears in your web browser. (4)

[17]

TOTAL: 100

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